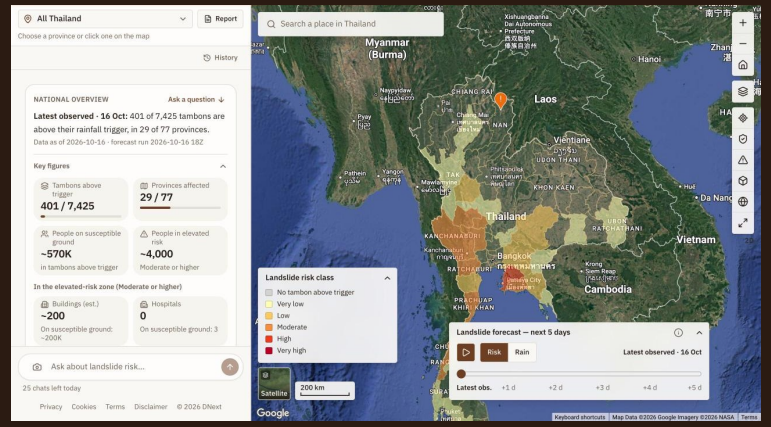
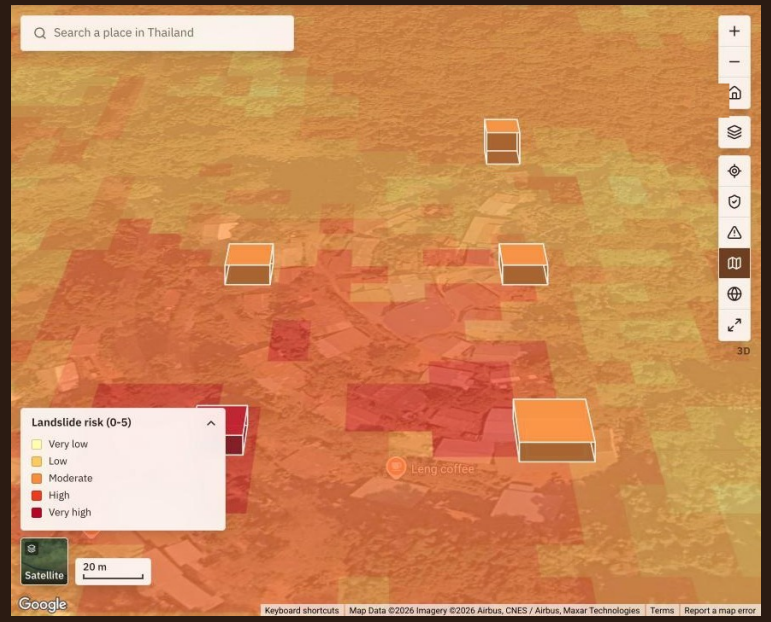
DNext · LandslideWatch TH

Landslide risk screening for Thailand, in plain language: where heavy rain is most likely to trigger landslides, and who and what sits in the way.

Theme: Sustainability & Social Impact
Category: Sustainability · Live: dnext-web-37ob55mcga-as.a.run.app





Heavy rain triggers landslides, and planners lack a fast local view

- Every rainy season, heavy rain sets off landslides on susceptible slopes across Thailand.
- The data that answers "how bad could it get here?" sits in separate systems: rainfall, slope susceptibility, population, buildings, roads. Combining it takes a specialist and time.
- People planning ahead need three answers quickly: which sub-districts are close to their rainfall trigger, who and what lies in the way, and where people can go.

7,425
tambons (sub-districts), each with its own rainfall thresholds

77
provinces covered, updated twice a day

30 days
of observed rain plus a 5-day forecast, compared with each tambon's own history

Who it helps, and how

Planners and local government

- Compare 150 mm with 300 mm before the rain arrives.
- See which sub-districts to check first.
- Export an editable PDF report in English or Thai.

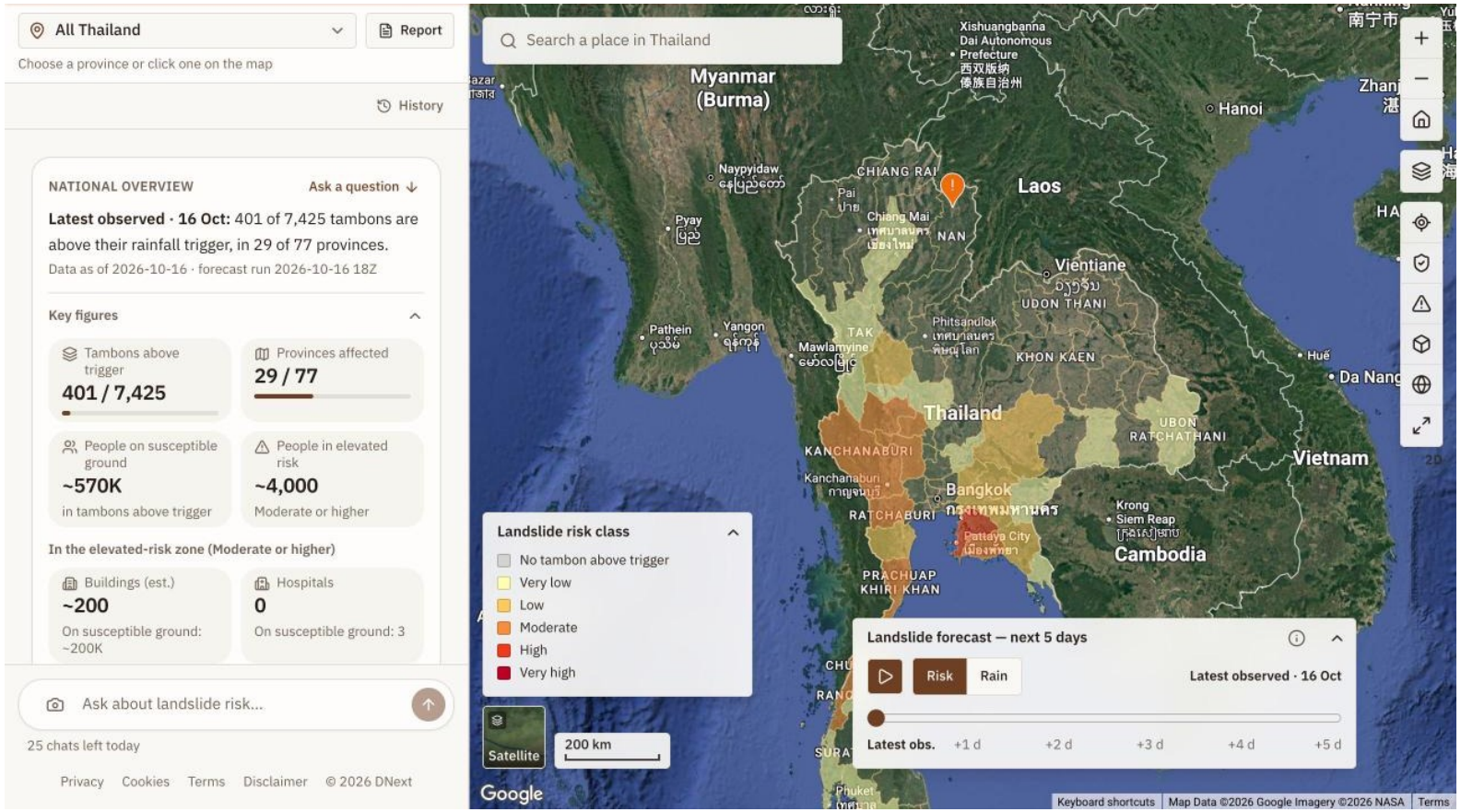
Responders, hospitals and schools

- See buildings, main roads, hospitals and schools inside the Moderate-or-higher zone.
- Follow a 5-day outlook and get an in-app alert when a watched province gets riskier.

Communities

- Report a landslide or blocked road with a photo that AI checks before it is shown.
- Find candidate safe places on low-risk ground and preview a route.

One screen for the whole country, today and for the next five days



Live screen, 16 October 2026: national overview with the 5-day forecast slider.

Risk by province

Latest observed risk, coloured by class, with a play button for the next five days.

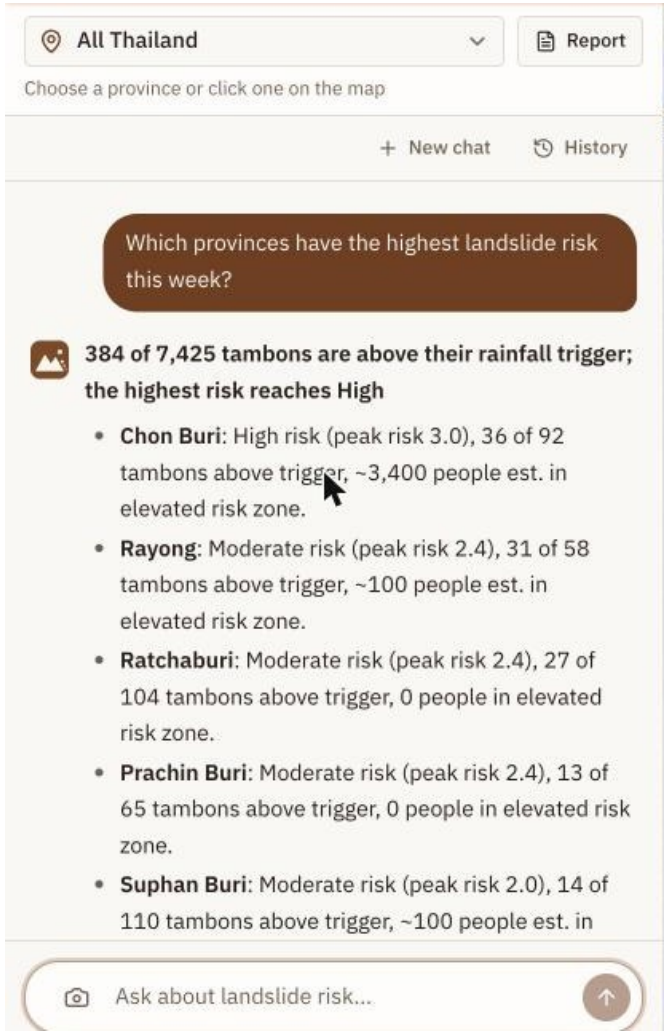
Key figures

401 of 7,425 tambons above their rainfall trigger in 29 of 77 provinces (16 Oct).

Reference layers

Susceptibility, vulnerability, exposure and population, each switchable.

Ask in plain language, in English or Thai



What if 250 mm falls on Nan in 3 days?

Risk map, tambon table, people, buildings, roads, hospitals and schools.

Which provinces have the highest landslide risk this week?

A ranked list and table, with the data date and forecast run time.

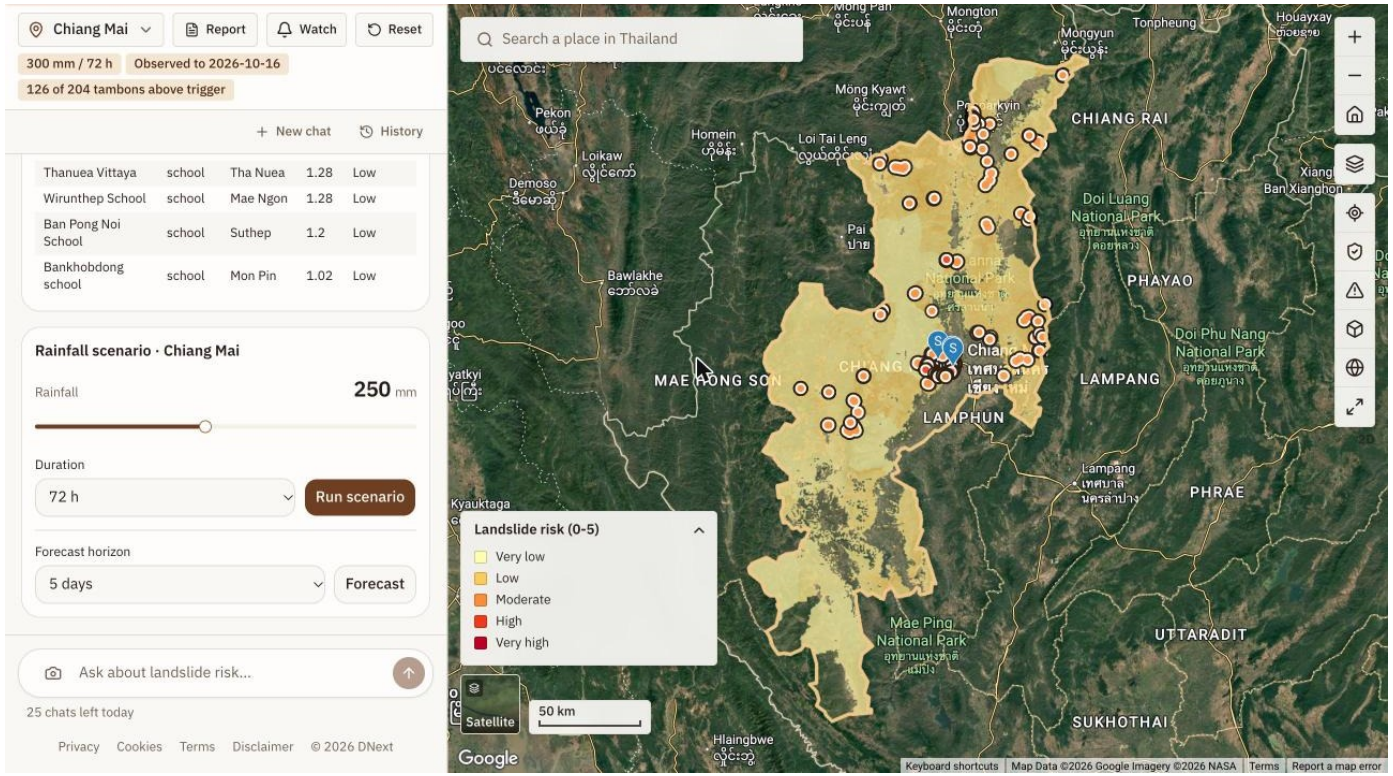
How many schools and hospitals are at risk in Chon Buri?

Counts for the country, a province, a district or one tambon, plus named facilities.

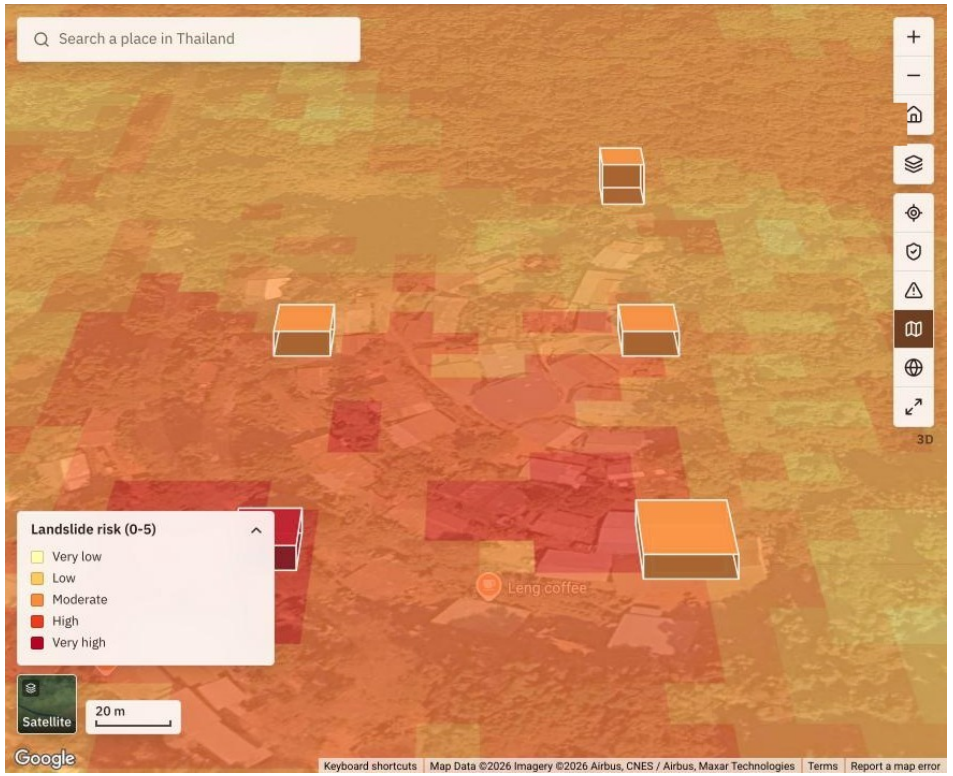
And in Chiang Mai? / What about next week?

Follow-ups keep the place, the time horizon and the risk class.

See what is exposed: buildings, roads, hospitals and schools



300 mm in 72 h, Chiang Mai: 3,424 buildings and 61.6 km of main road sit in the Moderate-or-higher zone. Each building is a point with a dark ring and white outline so thousands stay readable.



Street-level 3D: each affected footprint is raised by its estimated height. A "height exaggeration" control (x1 to x5) makes low rural buildings easier to see.

Every number is calculated, never written by the model

$$\text{risk} = \text{trigger} \times \text{susceptibility} \times \text{vulnerability} \times \text{exposure} \div 125$$

Trigger

A tambon's 30 days of rain (observed satellite rain plus the GFS forecast, or the what-if amount) against that tambon's own rainfall thresholds.

Susceptibility

A landslide susceptibility map at 12 m, classed 1 to 5.

Vulnerability

A national vulnerability index, classed 1 to 5.

Exposure

A national exposure index, classed 1 to 5.

Data: JAXA GSMaP and NOAA GFS rainfall · ADPC susceptibility, vulnerability and exposure layers through Google Earth Engine · WorldPop 2020 · GHSL built-up area · Google Open Buildings · OpenStreetMap. A tambon counts as landslide-prone only with at least 1 km² and 2% susceptible ground, so flat cities show no false risk.

Gen AI at the core: an agent that calls tools and never invents figures

Question
English or Thai

→

Gemini agent
Vertex AI · Agent Development Kit

→

10 tools
scenario, forecast, rankings, exposure, explain a point, briefing

→

Computed data
daily files and Earth Engine

→

Answer
table, map, short text

The model chooses tools; the tools compute

Every figure, name and class in an answer comes from a tool result. Ranking and "how many at risk" questions read the daily files directly, and a what-if is run only when the user gives a rainfall amount.

Follow-ups keep the conversation

Place, time horizon and risk class carry over. A repeated place name such as "Bang Phra" gets one clarifying question instead of a guess.

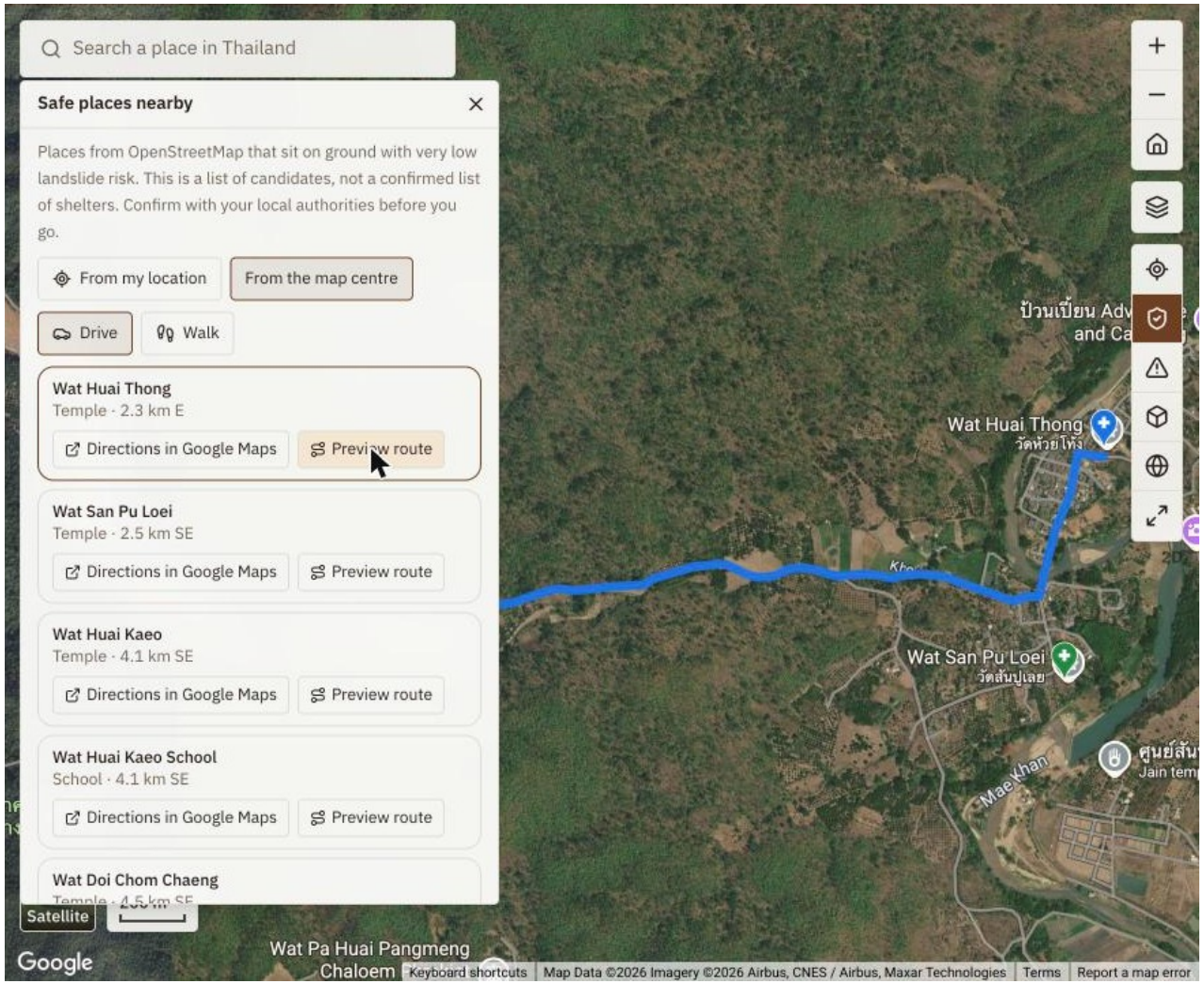
Gemini also reads photos

Community photos are checked for landslide, blocked road or slope damage, then compared with the area's data before anything is shown. The photo is never stored.

Tested, not assumed

A 20-case evaluation set checks tool choices, plus 260+ automated tests. The closing caveat is fixed wording in English and Thai.

Communities can report, and find somewhere safer to go



Report what you see

A photo of a landslide, blocked road or slope damage. Gemini reads it and the data for the place is checked. Only reports where both agree are shown, for 48 hours (7 days for slope damage). Three flags hide a report.

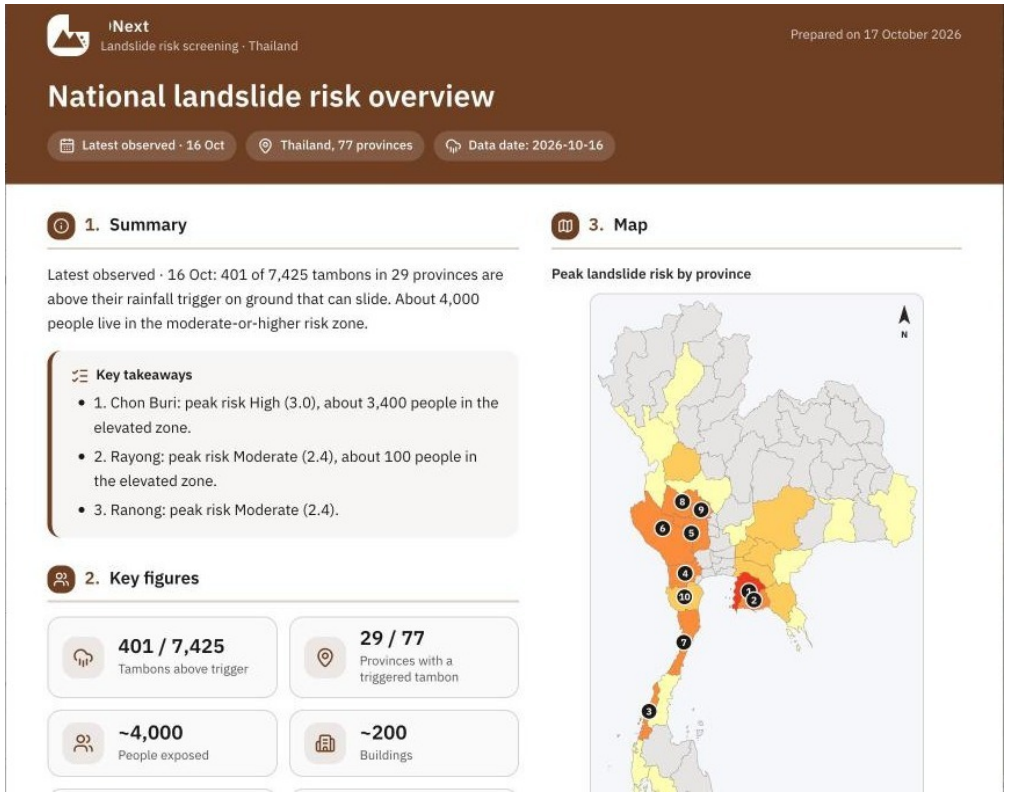
Find candidate safe places

26,425 named places from OpenStreetMap (shelters, community buildings, schools, temples, hospitals) on ground that stays at very low risk.

Preview a route

Google Routes API, ranked by how little of each route follows risky main-road sections. Never a guarantee of safety.

Reports to share, and alerts when risk rises



The report editor: click any text to edit before exporting.

An editable report in one click

Maps, key figures, charts, priority areas and general safety guidance for the result on screen or the national overview. Click any text to edit, then save as PDF in English or Thai.

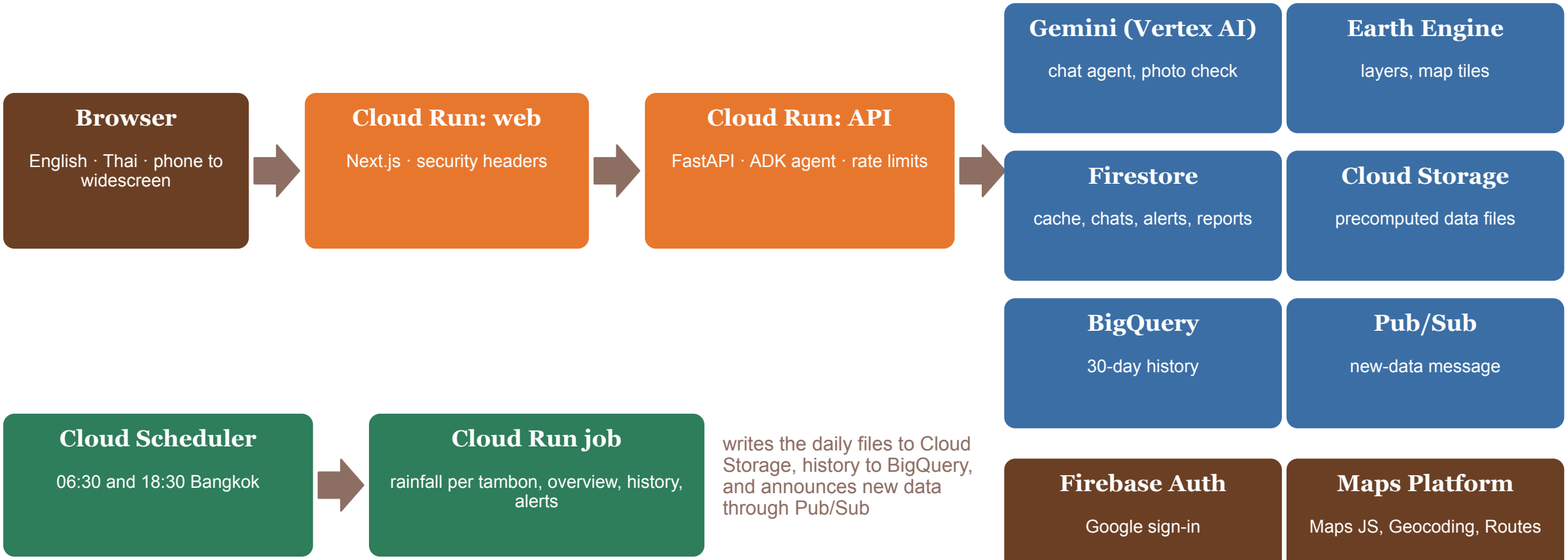
Watch a province

Signed-in users watch up to 10 provinces and get an alert in the app when the peak risk of one rises. Alerts are notices, not warnings.

History and a daily rhythm

A Cloud Run job runs at 06:30 and 18:30 Bangkok time, refreshes rainfall, appends the day to BigQuery (30-day trend) and announces new data through Pub/Sub.

Built on Google Cloud, serverless end to end



Also: Secret Manager, Artifact Registry, Cloud Build and Cloud Logging for deployment and operations.

Feasible today, and built to scale and to last

Fast by design

Heavy statistics are precomputed twice a day. The 5-day national overview answers in about 0.2 s; a scenario in roughly 1 to 8 s.

Elastic and shared

Cloud Run scales with traffic. Simultaneous requests share one computation, results are cached in Firestore and in memory, and stale data is served while it refreshes.

Cost-controlled

Per-account daily limits (30 chats, 200 runs, 40 route previews), a daily route cap, photos analysed once and never stored, records that expire on their own.

Secure by default

Google sign-in, per-caller rate limits, security headers, dependency audits in CI, no keys in the repository or the images.

Portable: replace the susceptibility, vulnerability and exposure layers and the same app covers another country or another rainfall-triggered hazard.

Honest about limits, clear about what comes next

What DNext is not

- A screening tool for planning, not a warning or an evacuation order.
- Not calibrated for single slopes or villages; classes are relative, not probabilities.
- What-if rain is assumed uniform across the area.
- Buildings and people are rounded estimates; OpenStreetMap can be incomplete.
- Safe places are candidates. Community reports are checked by AI, not verified.

Next steps

- Calibrate the thresholds against historical landslide records.
- Pilot with provincial planning teams and collect their feedback.
- Add flood and other rainfall-triggered hazards.
- Extend to other ASEAN countries by swapping the national layers.
- Short summaries for messaging apps, for places with weak connections.

DNext does not replace local knowledge. It helps people decide where to look first.

Try it live

https://dnext-web-37ob55mcga-as.a.run.app

1 Open the page

The national map and the 5-day forecast show without signing in. Press play.

2 Sign in with Google

Needed for chat, scenarios, routes and reports (per-account daily limits apply).

3 Try a question

"What if 250 mm falls on Nan in 3 days?" · "Which provinces have the highest risk this week?" · "How many schools are at risk in Chon Buri?"

Theme: Sustainability & Social Impact · Category: Sustainability · Google Cloud: Gemini on Vertex AI, Agent Development Kit, Earth Engine, Cloud Run, Firestore, BigQuery, Pub/Sub, Cloud Storage, Firebase Authentication, Google Maps Platform